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UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING

DATABASE SYSTEMS (CO2013)

Assignment 1 Report

Shopee E-Commercial Marketplace Platform

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Class CC05

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Member list & Workload

No.	Fullname	Student ID	Problems	% effort	% done
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3	Mai Thiện Ngôn	2352819	Entities and Attributes Design Semantic Constraints Mapping Description	20%	100%
3	Đoàn Vĩ Tiến	2353175	Users and Main Functions EERD Design Mapping Schema	20%	100%
5	Ngô Minh Thuận	2353153	Collect and organize documents EERD Design Mapping Schema	20%	100%

Table 1: Member list & workload

1 Analyze and Describe the Data Requirements

1.1 Research related system

1.1.1 Overview of Shoppe Vietnam

After reviewing Vietnam's leading marketplaces, we select Shopee.vn *(https://shopee.vn)* as the target of analysis. Shopee operates under a **Marketplace model** that integrates both **C2C (Customer-to-Customer)** and **B2C (Business-to-Customer)** business structures.

Shopee supports a wide range of essential functions that are typical of large-scale e-commerce platforms. Users can register and manage personal accounts, create and manage online stores, browse through categorized products, add items to their shopping carts, place orders, and complete payments via multiple payment methods. The platform also enables real-time order tracking and provides features such as product ratings, customer reviews, promotions, discount vouchers, and loyalty programs. These features ensure that the platform delivers a convenient, interactive, and secure online shopping experience for both buyers and sellers.

1.1.2 Business Process Analysis

When users visit the Shopee homepage, they can register for an account. Either as a buyer by default and then the user will be able to apply to be a seller.

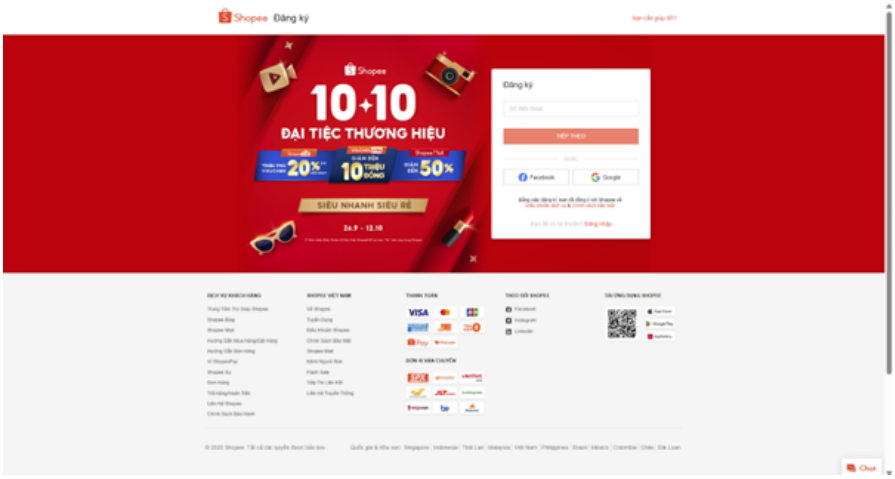

Figure 1: Shopee User Login Interface

Sellers can manage their online shops, list products, and oversee their inventory directly through the Shopee platform. To support new and existing sellers, Shopee also offers a variety of useful educational resources, including videos, articles, and tutorials—making it easier for sellers to get started and succeed on the platform.

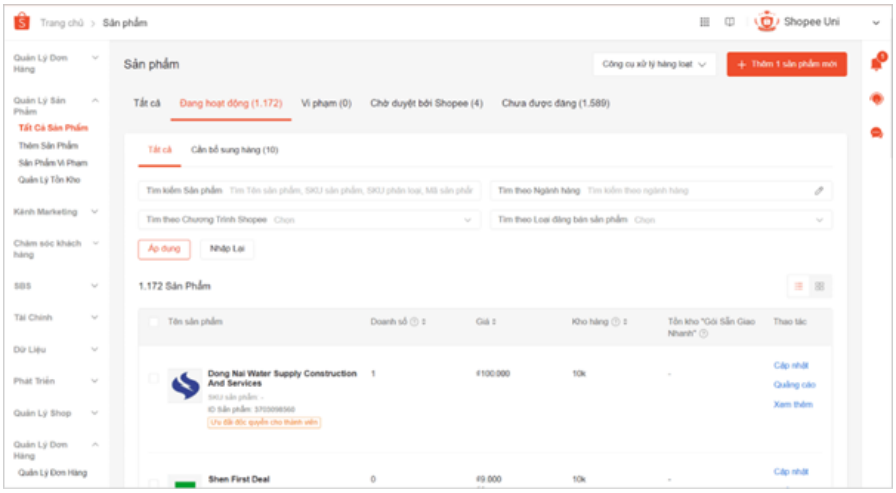

Figure 2: Shopee Seller Center

The buyer goes to the shopee main page, then use filters or the search bar to browse products . If the buyer chooses a product, there will be some variables for the buyer to choose from 1 product (take the shirt for example, there will be size, color, material,...). Then the product will either be transferred to the shopping cart if the player chooses "Add to shopping cart" or be transferred directly to the confirmation page. The buyer can apply the vouchers, then place the order.

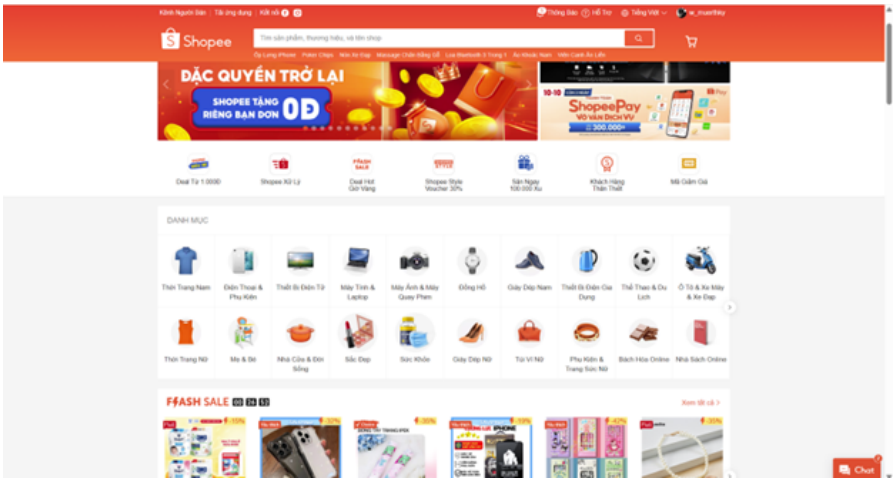

Figure 3: Shopee Homepage Interface

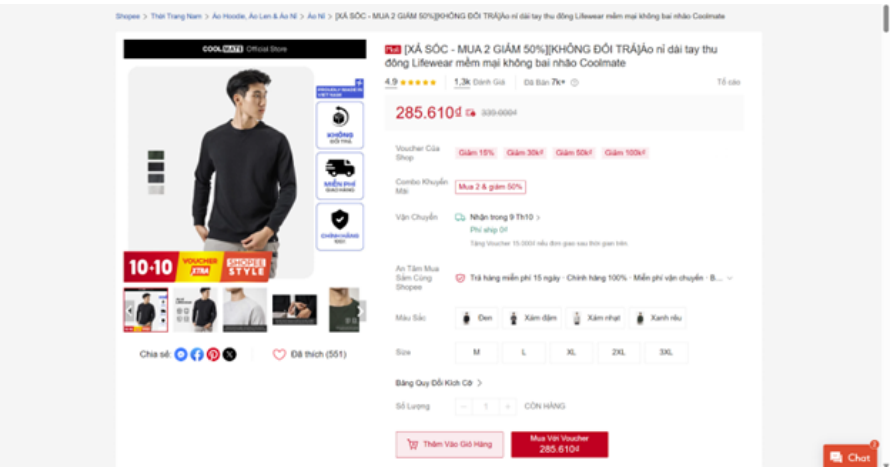

Figure 4: Buyer Product Interface

After the order is confirmed, Shopee validates the payment methods, locks inventory and sends delivery requests to the logistics team. The seller prepares the product and hands them over to the shipping team. The system generates an order record containing buyer information, seller details, product list, total cost, shipping address, and order status. The system continuously updates the order status (“Processing”, “Shipping”, “Delivered”) so that the buyer can track the order.

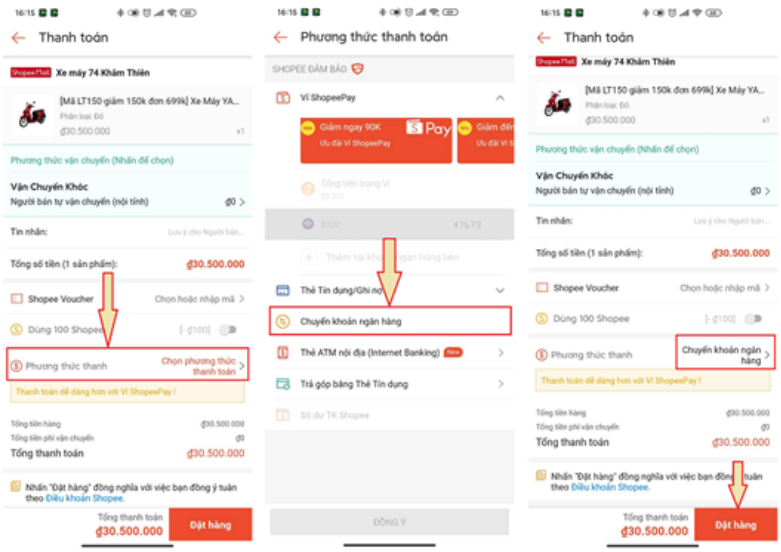

Figure 5: Shopee Order Confirmation Interface

After successful delivery and buyer confirmation, funds are released to sellers, minus commission and service fees. Returns and refund workflows are initiated when a buyer is unsatisfied or the product is defective. Throughout the cycle, buyers can rate and review both products and sellers. Sellers can access dashboards for analytics, while Shopee administrators manage disputes, monitor activity, and schedule platform-wide promotions such as 11.11 or 9.9 sales.

1.2 Proposed System

1.2.1 Users and Main Functions

The proposed application is a **comprehensive e-commerce platform** designed to facilitate online buying and selling activities between customers and merchants. The system aims to provide a seamless, secure, and user-friendly environment where users can browse, purchase, and manage products and orders efficiently. It incorporates essential modules such as user management, product cataloging, order and payment processing, shipping and delivery tracking, dispute resolution, and review management, ensuring a complete and integrated online shopping experience.

The platform supports three primary types of users: **buyers**, **sellers**, and **administrators**. **Buyers** are regular customers who register on the platform to browse through various product categories, compare prices, and make purchases. They can add products to their order, proceed with secure payments using multiple payment methods, and track their orders in real-time. Buyers also have the option to return products, request refunds, and provide ratings and reviews for the items they purchase, contributing to transparency and trust within the platform.

Sellers, on the other hand, are individuals or businesses who register to create and manage their online shops. They are responsible for uploading product information, including descriptions, stock quantities, and price variants. Sellers can also set up promotional campaigns, manage discounts, and monitor sales performance through the platform's analytics features. They handle order fulfillment, arrange shipments, and respond to customer inquiries or disputes to ensure a satisfactory purchasing experience.

The third user type, the **administrator**, oversees the overall system operations. Admins manage user accounts, verify seller registrations, and ensure that transactions and listings comply with platform policies. They are also responsible for handling disputes and refunds, managing promotional campaigns, and monitoring the overall health of the marketplace. Through the admin panel, they can review reports, manage payouts to sellers, and take corrective actions against fraudulent or suspicious activities.

The core functions of the system revolve around its ability to manage users, products, orders, and transactions efficiently. The **user management module** handles account registration, authentication, and profile updates. The product management module enables sellers to add, update, and categorize products with ease. The **order and payment module** processes customer purchases, ensures secure payment transactions, and manages order statuses throughout the fulfillment cycle. Additionally, the **promotion system** allows both sellers and admins to create discount campaigns that encourage customer engagement and boost sales. The **review and rating module** fosters trust by allowing buyers to share their experiences, while the dispute and refund module provides a structured way to resolve item-related issues. Finally, **the shipment tracking feature** ensures that customers can monitor delivery progress from dispatch to receipt.

1.2.2 Data Requirements (Attributes, Entities and Relationships)

The first requirement of our marketplace system is to keep track of all **Users**. Each **User** has a profile containing a unique *UserID*, username, password, email, phone number, address, timestamp when registers, user type, status and full name including first name and last name. Based on the user type as well as the role in the system, **Users** are separated into **Buyers**, **Sellers** and **Admins(administrator)**.

Buyers are the end users of the system who buy products posted by the **Sellers**, make a payment and review the products. The system records the Buyers' total orders, default payment method and loyalty points.

Besides the Buyers, **Sellers** are also the end users of the system who post their products to the system to sell to the Buyers. Each Seller's shop name, shop description, total number of products and rating will be recorded by the system.

Admins are the employees who manage the operation of the system. They have additional attributes such as role and hired date.

The **Product** entity stores general information about each product available in the system. Each product has a unique *ProductID* along with its name, description, category, status, average rating, total number of reviews, the timestamp added and some images about the product. Products can also be related to other products through a recursive relationship called RelateTo, which represents associations such as similar items, complementary goods, or recommended alternatives within the catalog. A product can have no or multiple product variants, which represent specific variations of the corresponding product such as size, color, price. Each **ProductVariant** is uniquely identified by a *VariantID* and includes detailed attributes like variant name, color, size, price, stock quantity and status. While a product may exist without any variants, every product variant must belong to exactly one product.

The **Order** entity represents a customer's purchase order created within the system. Each Order is uniquely identified by an *OrderID* and stores general information such as order price, status, the timestamp when the order was placed and the tax amount which can be derived from order price. An **Order** reflects the customer's intent to buy one or more products, serving as a container that groups all selected items into a single purchase record. Each **Order** includes multiple order items, representing the specific products and quantities the customer requested.

The **OrderItem** entity stores detailed information about each product variant included in an order. Each order item includes a unique *OrderItemID* and quantity of the product variant corresponding to it. Every order item refers to exactly one product variant, defining which specific variation of a product was ordered and in what quantity. While an order may contain multiple order items, each order item is always associated with exactly one order and one product variant.

The **OrderPayment** entity represents the payment transaction made by a buyer to the marketplace system (admin) for a specific order. Each **OrderPayment** is uniquely identified by a *PaymentID* and is associated with one specific buyer, order and admin. It has other details such as payment information, method, status, amount of payment and timestamp when payment is executed.

The **PayoutPayment** entity manages disbursement transactions from the platform administrators to sellers after successful order completion. Each payout record is uniquely identified by a *PayoutID* and includes details such as payout information, payout amount, payout method, status, and the time payout transaction performed. A payout is issued by one admin and may involve one or multiple sellers depending on whether the payment is processed individually or in batch. Each payout corresponds to one settled order, ensuring that sellers receive payment accurately and the transaction history is properly recorded for financial tracking and transparency.

The **Dispute** entity captures complaint cases submitted by the buyers to the admins regarding issues with purchased orders, such as damaged, missing, or incorrect items. Each Dispute has a unique *DisputeID*, dispute reason, status, decision of the admin and the timestamps when dispute is created and resoluted. A dispute is related to one specific order, one buyer, one admin who handles the case and a buyer can create multiple disputes to the admin over time. Besides, each dispute is raised against one or more sellers who join in the same order and each seller can receive multiple disputes.

The **RefundPayment** entity represents the refund transaction initiated by the admins to return money to buyers when a dispute is resolved in favor of the buyer. Each refund record is uniquely identified by a *RefundID* and contains details such as refund information, refund method, refund amount, transaction status and the timestamp when the transaction is executed. A refund payment is always linked to one dispute, one buyer, and one admin, reflecting the financial resolution process following a complaint. While not all disputes result in refunds, each refund payment is associated with exactly one dispute.

The **Return** entity records product return activities where the buyer sends the purchased items back to the seller after a dispute has been approved by the admin. Each return is associated with one dispute, has a unique *ReturnID* and includes attributes such as return reason, return method, tracking code and the timestamps when buyer returns and seller receives the products. While not all disputes result in returns, each return is associated with exactly one dispute.

The **Shipment** entity represents the delivery process of an order after it has been confirmed and paid. Each shipment record is uniquely identified by a *ShipmentID* and includes attributes such as tracking code, shipping provider, shipping fee, shipment status, pickup date, estimated delivery date, and actual delivery date, receiver name, receiver phone, shipping address specify the delivery destination details and the timestamps when shipping provider picks up the order, estimated and actual time the order is delivered to the buyer. A shipment is associated with exactly one order, but each order can generate at most one shipment record.

The **Review** entity stores feedback information submitted by buyers after completing an order. Each review record is uniquely identified by a *ReviewID* and includes attributes such as rating, review content and the timestamp when the review is created. Every review from a buyer is associated with exactly one product variant and one order, ensuring that feedback is linked to a verified purchase. While a buyer can create multiple reviews over time, each review corresponds to a specific product variant from a specific order, helping maintain credibility and accuracy of product evaluations.

The **Promotion** entity represents discount campaigns or special offers created by either sellers or administrators in the marketplace system. Each promotion record is uniquely identified by a *PromotionID* and includes attributes such as promotion name, description, discount type, quantity, requirements, status and the timestamps when the promotion starts and ends. The creator of a promotion, referred to as the **PromotionCreator**, can be either a seller (offering shop-level discounts) or an admin (offering platform-wide vouchers). A promotion can be applied to multiple orders, while each order may use one or more applicable promotions depending on the campaign rules.

1.3 Semantic Constraints

Order Status Progression Constraint: An order must follow a specific lifecycle sequence. For instance, an order cannot move directly from "pending" status to "delivered" status without first passing through "processing" and "shipped" states.

Review Eligibility Constraint: A user can only create a review of a ProductVariant if they have at least one order containing that ProductVariant with status "Delivered/Completed". This ensures only customers who have actually received the specific variant can review it.

Inventory Consistency Constraint: The total quantity of ProductVariants ordered cannot exceed available stock, and inventory must be updated when orders are confirmed and restored when orders are cancelled.

Payment-Order Amount Matching Constraint: The payment amount must exactly match the order total (sum of order items + shipping + taxes - discounts). No partial payment is allowed at the moment.

Refund Authorization Constraint: Refunds can only be issued for paid orders, cannot exceed the original payment amount, and must be requested within the allowed time window after the order status is changed to “Delivered”.

Order status: Must be one of the following: Placed, Preparing to Ship, In Transit, Out for Delivery, Delivered, Completed, Disputed, Return Processing, Return Completed, Refunded, Cancelled.

Payment status: Must be one of the following: Unpaid, Processing, Processed, Failed, Refunded.

Refund Status: Must be one of the following: Created, Processing, Completed, Failed.

Dispute Status: Must be one of the following: Created, Under Review, Awaiting Seller Response, Awaiting Buyer Information, Escalated, Closed

Payout Status: Must be one of the following: Created, Processing, Completed, Failed

Product Category Assignment Constraint: Every product must be assigned to at least one valid category that exists in the category hierarchy, and a product cannot be assigned to categories that are marked as inactive or deleted.

Shipping Address Requirement Constraint: Before an order can transition to "shipped" status, it must have a complete and validated shipping address associated with it, including all required fields like street address, city, postal code, and country.

User Role-Based Action Constraint: Certain operations can only be performed by users with specific roles. For example, only users with "buyer" or “admin” roles can delete products, modify prices, or access customer personal information, while regular customers can only manage their own orders and reviews.

Rating Value Range Constraint: Any review rating must fall within the defined valid range, typically between 1 and 5 stars.

Order Cancellation Time Constraint: A customer can cancel an order only if it hasn't yet reached certain status milestones. For instance, once an order status becomes "shipped" or "delivered," it cannot be cancelled by the customer and must go through a return/refund/dispute process instead.

2 EERD Design

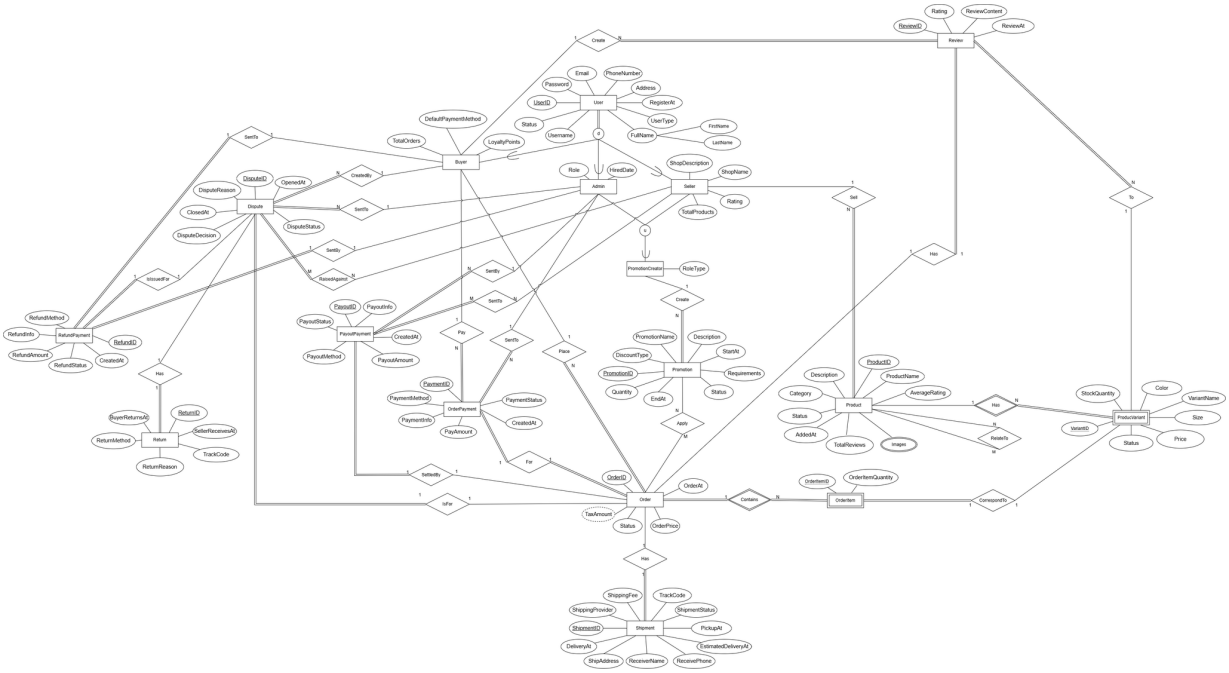

Figure 6: Entity Relationship Diagram

3 Mapping to Database Schema

3.1 Data model mapping

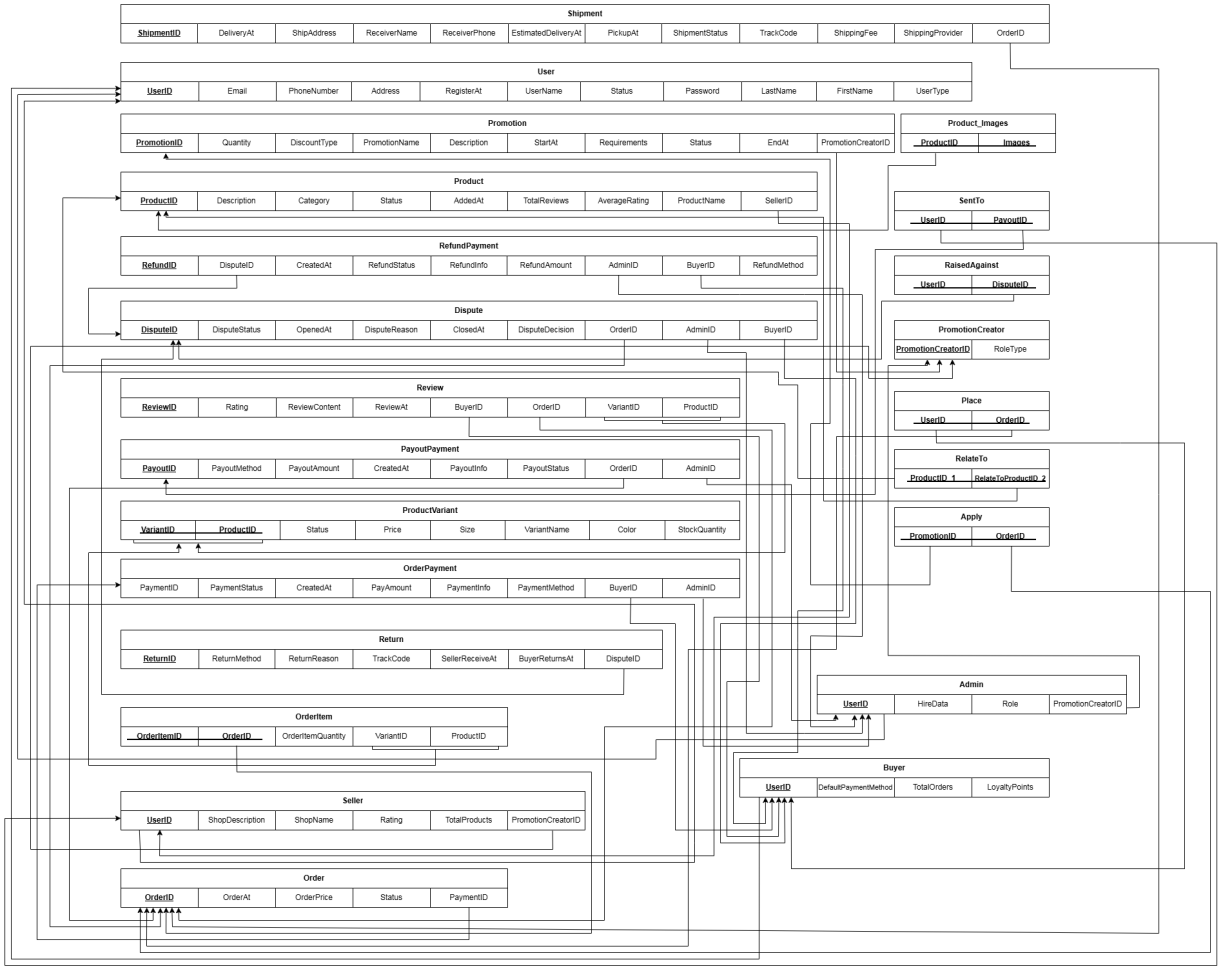

Figure 7: Data Model Mapping

3.2 Description

Attribute	Data type	Definition	Specification
UserID	INT	Unique identifier for each user	PRIMARY KEY, AUTO_INCREMENT
UserName	VARCHAR(50)	User's chosen login name	NOT NULL, UNIQUE
Password	VARCHAR(255)	Encrypted password for authentication	NOT NULL
FirstName	VARCHAR(100)	First name of the user	NOT NULL
LastName	VARCHAR(100)	Last name of the user	NOT NULL
Email	VARCHAR(100)	Email address of the user	NOT NULL, UNIQUE
PhoneNumber	VARCHAR(10)	Contact number of the user	NULL
Address	VARCHAR(255)	Registered address of the user	NULL
RegisterAt	DATETIME	Date and time when user account was created	DEFAULT CURRENT _TIMESTAMP
UserType	ENUM('Admin', 'Buyer','Seller')	Indicates whether the user is an Admin, Buyer, or Seller	NOT NULL
Status	ENUM('Active', 'Suspended','Deleted')	Current status of the user account	DEFAULT 'Active'

Table 2: Description of **User** table

Attribute	Data type	Definition	Specification
UserID	INT	References the corresponding UserID from the User entity	NOT NULL, PRIMARY KEY, FOREIGN KEY REFERENCES User(UserID)
Role	VARCHAR(50)	The role or responsibility level of the admin	NOT NULL
HiredData	VARCHAR(255)	Administrative or employment-related data (internal HR data or record)	NULL
Promotion CreatorID	INT	References PromotionCreator if the Admin creates promotions	FOREIGN KEY REFERENCES PromotionCreator (PromotionCreatorID)

Table 3: Description of **Admin** table

Attribute	Data type	Definition	Specification
UserID	INT	References the corresponding UserID from the User entity	NOT NULL, PRIMARY KEY, FOREIGN KEY REFERENCES User(UserID)
LoyaltyPoints	INT	Total reward or loyalty points accumulated by the buyer	DEFAULT 0
DefaultPayment Method	VARCHAR(50)	Preferred payment method used by the buyer	NULL
TotalOrders	INT	Total number of orders placed by the buyer	DEFAULT 0

Table 4: Description of **Buyer** table

Attribute	Data type	Definition	Specification
UserID	INT	References the corresponding UserID from the User entity	NOT NULL, PRIMARY KEY, FOREIGN KEY REFERENCES User(UserID)
ShopName	VARCHAR(100)	Name of the seller's shop displayed on the platform	NOT NULL
ShopDescription	TEXT	Description or introduction of the seller's shop	NULL
Rating	DECIMAL(2,1)	Average rating score of the seller's shop	DEFAULT 0.0
TotalProducts	INT	Total number of products currently listed by the seller	DEFAULT 0
Promotion CreatorID	INT	References PromotionCreator if the Seller creates promotions	FOREIGN KEY REFERENCES PromotionCreator (PromotionCreatorID)

Table 5: Description of **Seller** table

Attribute	Data type	Definition	Specification
Promotion CreatorID	INT	Unique identifier for each promotion creator	PRIMARY KEY, AUTO_INCREMENT
RoleType	ENUM('Admin','Seller')	Defines whether the promotion creator is an Admin or Seller	NOT NULL

Table 6: Description of **PromotionCreator** table

Attribute	Data type	Definition	Specification
PromotionID	INT	Unique identifier for each promotion	PRIMARY KEY, AUTO_INCREMENT
Promotion CreatorID	INT	References the creator (Admin or Seller) from PromotionCreator	NOT NULL, FOREIGN KEY REFERENCES PromotionCreator (PromotionCreatorID)
Promotion Name	VARCHAR(100)	Name or title of the promotion	NOT NULL,
Description	TEXT	Details or explanation of the promotion	NULL
DiscountType	ENUM('Percentage', 'FixedAmount', 'FreeShipping','Voucher')	Type of discount offered in the promotion	NOT NULL
StartAt	DATETIME	Date and time when the promotion becomes active	NOT NULL
EndAt	DATETIME	Date and time when the promotion ends	NOT NULL
Requirements	VARCHAR(255)	Conditions or eligibility rules to apply the promotion	NULL
Quantity	INT	Total number of times the promotion can be used	DEFAULT 0
Status	ENUM('Active', 'Expired','Upcoming')	Current status of the promotion	DEFAULT 'Upcoming'

Table 7: Description of **Promotion** table

Attribute	Data type	Definition	Specification
OrderID	INT	Unique identifier for each order placed in the system	PRIMARY KEY, AUTO_INCREMENT
OrderAt	DATETIME	Timestamp recording exactly when the customer placed this order	NOT NULL
OrderPrice	DECIMAL(10,2)	Total monetary value of all items in the order before taxes, shipping, or discounts	NOT NULL, CHECK (OrderPrice >= 0)
Status	VARCHAR(50)	Current state of the order in its lifecycle (Stated in the Semantic Constrant)	NOT NULL
PaymentID	INT	Links to the payment transaction that was used to pay for this order	NULL, FOREIGN KEY REFERENCES Order Payment(PaymentID)

Table 8: Description of **Order** table

Attribute	Data type	Definition	Specification
PaymentID	INT	Unique identifier for each payment transaction attempt	PRIMARY KEY, AUTO_INCREMENT
Payment Status	VARCAHR(20)	Current state of the payment process (Stated in Semantic Constrant)	NOT NULL
CreatedAt	DATETIME	Timestamp when the payment record was first created in the system	NOT NULL
PayAmount	DECIMAL(10,2)	The actual amount of money that was charged or attempted to be charged	NOT NULL, CHECK (PayAmount > 0)
PaymentInfo	VARCHAR(200)	Additional payment details such as transaction ID from payment gateway, last four digits of card, or confirmation code	NULL
Payment Method	VARCHAR(50)	The payment mechanism used (e.g., Credit Card, Debit Card, MobileWallet, Bank Transfer, Cash on Delivery)	NOT NULL
BuyerID	INT	Identifier of the buyer who is making this payment	NOT NULL, FOREIGN KEY REFERENCES Buyer(UserID)
AdminID	INT	Identifier of the buyer who manually processed this payment (Receive to Escrow)	NULL, FOREIGN KEY REFERENCES Admin(UserID)

Table 9: Description of **OrderPayment** table

Attribute	Data type	Definition	Specification
Order ItemID	INT	Unique identifier for each individual item withih an order	PRIMARY KEY, AUTO_INCREMENT
OrderID	INT	Links this item to its parent order	NOT NULL, FOREIGN KEY REFERENCES Order(OrderID)
OrderItem Quantity	INT	Number of units of this product variant being purchased	NOT NULL, CHECK (OrderItemQuantity > 0)
VariantID	INT	Identifies the specific product variant being purchased	NOT NULL, FOREIGN KEY REFERENCES ProductVariant(VariantID)
ProductID	INT	Identifies the main product this item belongs to	NOT NULL, FOREIGN KEY REFERENCES Product(ProductID)

Table 10: Description of **OrderItem** table

Attribute	Data type	Definition	Specification
RefundID	INT	Unique identifier for each refund transaction processed	PRIMARY KEY, AUTO_INCREMENT
DisputeID	INT	Links this refund to the dispute case that triggered it, if applicable	NULL, FOREIGN KEY REFERENCES Dispute(DisputeID)
CreatedAt	DATETIME	Timestamp when the refund was initiated or created in the system	NOT NULL
Refund Status	VARCHAR(20)	Current state of the refund process (e.g., Pending, Processing, Completed, Failed)	NOT NULL
Refund Info	VARCHAR(200)	Additional details about the refund such as transaction reference, reason code, or processing notes	NULL
Refund Amount	DECIMAL(10,2)	The monetary amount being returned to the customer	NOT NULL, CHECK (RefundAmount > 0)
BuyerID	INT	Identifier of the buyer receiving this refund	NOT NULL, FOREIGN KEY REFERENCES Buyer(UserID)
AdminID	INT	Identifier of the admin who authorized or processed this refund	NULL, FOREIGN KEY REFERENCES Admin(UserID)

Table 11: Description of **RefundPayment** table

Attribute	Data type	Definition	Specification
DisputeID	INT	Unique identifier for each dispute case opened in the system	PRIMARY KEY, AUTO_INCREMENT
Dispute Status	VARCHAR(30)	Current state of the dispute resolution process (e.g., Open, Under Review, Awaiting Seller Response, Escalated, Resolved, Closed)	NOT NULL
OpenedAt	DATETIME	Timestamp when the dispute was first filled by the customer	NOT NULL
Dispute Reason	VARCHAR(100)	Category or classification of why the dispute was filed (e.g., Item Not Received, Item Not As Described, Damaged, Wrong Item)	NOT NULL
ClosedAt	DATETIME	Timestamp when the dispute was resolved and closed	NULL
Dispute Decision	TEXT	Detailed explanation of how the dispute was resolved, including actions taken and final determination	NULL
OrderID	INT	Links the dispute to the specific order being contested	NOT NULL, FOREIGN KEY REFERENCES Order(OrderID)
BuyerID	INT	Identifier of the buyer who filed the dispute	NOT NULL, FOREIGN KEY REFERENCES Buyer(UserID)
AdminID	INT	Identifier of the admin assigned to investigate this case	NULL, FOREIGN KEY REFERENCES Admin(UserID)

Table 12: Description of **Dispute** table

Attribute	Data type	Definition	Specification
PayoutID	INT	Unique identifier for each payout transaction to a seller	PRIMARY KEY, AUTO_INCREMENT
Payout Method	VARCHAR(50)	The mechanism used to send money to the seller (e.g., Bank Transfer, PayPal, Check, Wire Transfer)	NOT NULL
Payout Amount	DECIMAL(10,2)	The monetary amount being paid out to the seller	NOT NULL, CHECK (PayoutAmount > 0)
CreatedAt	DATETIME	Timestamp when the payout was initiated or scheduled	NOT NULL
Payout Info	VARCHAR(200)	Additional details about the payout such as transaction reference, bank account details (masked), or cofirmation codes	NULL
Payout Status	VARCHAR(20)	Current state of the payout process (e.g., Pending, Processing, Completed, Failed, Returned) or	NOT NULL
OrderID	INT	Links this payout to the specific order that generated the revenue being paid out	NOT NULL, FOREIGN KEY REFERENCES Order(OrderID)
SellerID	INT	Identifier of the seller receiving this payout	NOT NULL, FOREIGN KEY REFERENCES Seller(UserID)
AdminID	INT	Identifier of the admin who initiated or approved this payout	NULL, FOREIGN KEY REFERENCES Admin(UserID)

Table 13: Description of **PayoutPayment** table

UserID	INT	Identifier of the buyer who placed the order	PRIMARY KEY (composite) FOREIGN KEY REFERENCES Buyer(UserID)
OrderID	INT	Identifier of the order that was placed by this buyer	PRIMARY KEY (composite) FOREIGN KEY REFERENCES Order(OrderID)

Table 14: Description of **Place** table

ProductID_1	INT	Identifier of the primary product in the relationship	PRIMARY KEY (composite) FOREIGN KEY REFERENCES Product(ProductID)
RelateTo ProductID_2	INT	Identifier of the related product being suggested (e.g., for 'customers who bought this also bought' recommendations)	PRIMARY KEY (composite) FOREIGN KEY REFERENCES Product(ProductID), CHECK (ProductID_1 != RelateToProductID_2)

Table 15: Description of **RelateTo** table

PromotionID	INT	Identifier of the promotion/discount being applied to the order	PRIMARY KEY (composite) FOREIGN KEY REFERENCES Promotion(PromotionID)
OrderID	INT	Identifier of the order receiving the promotional discount	PRIMARY KEY (composite) FOREIGN KEY REFERENCES Order(OrderID)

Table 16: Description of **Apply** table

Attribute	Data type	Definition	Specification
ReturnID	INT	Unique identifier for each return record	PRIMARY KEY
ReturnMethod	VARCHAR(30)	The method used by the buyer to return the product (e.g., shipping, pickup)	NOT NULL
ReturnReason	VARCHAR(255)	The reason why the buyer returned the product	NOT NULL
TrackCode	VARCHAR(50)	The tracking code used to monitor the return shipment	UNIQUE
SellerReceiveAt	DATETIME	The date and time when the seller confirmed receiving the returned product	NULL
BuyerReturnsAt	DATETIME	The date and time when the buyer sent back the product	NULL
DisputeID	INT	References the dispute record that triggered the return process	NOT NULL, FOREIGN KEY REFERENCES Dispute(DisputeID)

Table 17: Description of **Return** table

Attribute	Data type	Definition	Specification
ReviewID	INT	Unique identifier for each review record	PRIMARY KEY
Rating	INT	Rating score given by the buyer for the purchased product	NOT NULL
Review Content	VARCHAR(500)	Text content of the buyer's feedback	NULL
ReviewAt	DATETIME	Date and time when the review was submitted	NOT NULL
UserID	INT	References the buyer who created the review	NOT NULL, FOREIGN KEY REFERENCES Buyer(UserID)
OrderID	INT	References the order that the review is associated with	NOT NULL, FOREIGN KEY REFERENCES Order(OrderID)
VariantID	INT	References the specific product variant being reviewed	NOT NULL, FOREIGN KEY REFERENCES ProductVariant(VariantID)
ProductID	INT	References the main product that the variant belongs to	NOT NULL, FOREIGN KEY REFERENCES ProductVariant(VariantID)

Table 18: Description of **Review** table

Attribute	Data type	Definition	Specification
ProductID	INT	Unique identifier for each product in the system	PRIMARY KEY
UserID	INT	References the seller who listed the product	NOT NULL, FOREIGN KEY REFERENCES Seller(UserID)
Product Name	VARCHAR(100)	The official name or title of the product	NOT NULL
Description	TEXT	Detailed description or information about the product	NULL
Category	VARCHAR(50)	The category to which the product belongs (e.g., Electronics, Fashion)	NULL
Status	VARCHAR(20)	Indicates the current status of the product (e.g., Active, Inactive)	NOT NULL
AddedAt	DATETIME	The date and time when the product was added to the system	NOT NULL
Total Reviews	INT	Total number of reviews that the product has received	DEFAULT 0
Average Rating	DECIMAL(2,1)	The average rating score calculated from user reviews	DEFAULT 0.0

Table 19: Description of **Product** table

Attribute	Data type	Definition	Specification
VariantID	INT	Partial key uniquely identifying a variant within a product	PRIMARY KEY (composite)
ProductID	INT	Identifier referencing the product this variant belongs to	PRIMARY KEY (composite) FOREIGN KEY REFERENCES Product(ProductID)
Variant Name	VARCHAR(100)	Name or label of the product variant (e.g., "Red 128GB")	NOT NULL
Color	VARCHAR(50)	Color attribute of the variant	NULL
Size	VARCHAR(50)	Size or dimension information of the variant	NULL
Price	DECIMAL(10,2)	Price of the specific product variant	NOT NULL
Stock Quantity	INT	Available stock quantity of the variant	NOT NULL
Status	VARCHAR(20)	Availability status (e.g., "Available", "Out of Stock")	DEFAULT 'Available'

Table 20: Description of **ProductVariant** table

Attribute	Data type	Definition	Specification
ShipmentID	INT	Unique identifier for each shipment record	PRIMARY KEY
OderID	INT	Identifier referencing the order associated with this shipment	NOT NULL, FOREIGN KEY REFERENCES Order(OrderID)
ShipAddress	VARCHAR(200)	Address where the order will be delivered	NOT NULL
Receiver Name	VARCHAR(100)	Name of the recipient of the shipment	NOT NULL
Receiver Phone	VARCHAR(15)	Contact phone number of the recipient	NOT NULL
Shipping Provider	VARCHAR(50)	Name of the shipping or delivery service provider	NOT NULL
ShippingFee	DECIMAL(10,2)	Cost charged for the delivery service	DEFAULT 0.00
Shipment Status	VARCHAR(30)	Current shipment status (e.g., "Pending", "Shipped", "Delivered", "Returned")	DEFAULT 'Pending
TrackCode	VARCHAR(50)	Tracking code provided by the delivery service	UNIQUE
PickupAt	DATETIME	Timestamp when the package was picked up for delivery	NULL
Estimate DeliveryAt	DATETIME	Expected delivery date/time of the order	NULL
DeliveryAt	DATETIME	Actual date/time when the order was delivered	NULL

Table 21: Description of **Shipment** table

ProductID	INT	Identifier referencing the product associated with the image	PRIMARY KEY (composite) FOREIGN KEY REFERENCES Product(ProductID)
ImageURL	VARCHAR(255)	File path or URL of the product image stored in the system	PRIMARY KEY (composite)

Table 22: Description of **Product_Images** table

UserID	INT	Identifier referencing the seller who receives the payout	PRIMARY KEY (composite) FOREIGN KEY REFERENCES Seller(UserID)
PayoutID	INT	Identifier referencing the payout transaction sent to the seller	PRIMARY KEY (composite) FOREIGN KEY REFERENCES PayoutPayment(PayoutID)

Table 23: Description of **SentTo** table

UserID	INT	Identifier referencing the seller who is involved in the dispute	PRIMARY KEY (composite) FOREIGN KEY REFERENCES Seller(UserID)
DisputeID	INT	Identifier referencing the dispute raised against the seller	PRIMARY KEY (composite) FOREIGN KEY REFERENCES Dispute(DisputeID)

Table 24: Description of **RaisedAgainst** table